listing into it. You'll need to add the following imports at the top of the file:

```
import COSE.*;
import com.upokecenter.cbor.CBORObject;
import org.bouncycastle.jce.provider.BouncyCastleProvider;
import java.nio.*;
import java.security.*;
```

Listing 13.8 Deriving the sender and recipient keys

```
private static Key deriveKey(Key hkdfKey, byte[] id,
    byte[] idContext, AlgorithmID coseAlgorithm)
        throws GeneralSecurityException {

    int keySizeBytes = coseAlgorithm.getKeySize() / 8;
    CBORObject context = CBORObject.NewArray();    #A
    context.Add(id);    #A
    context.Add(idContext);    #A
    context.Add(coseAlgorithm.AsCBOR());    #A
    context.Add(CBORObject.FromObject("Key"));    #A
    context.Add(keySizeBytes);    #A

    return HKDF.expand(hkdfKey, context.EncodeToBytes(),
#B
            keySizeBytes, "AES");    #B
}
```

#A The context is a CBOR array containing the ID, ID context, algorithm identifier, and key size.